

Understanding Timing Report in OpenSTA

INTRODUCTION:

A timing report is a crucial output file generated by a Static Timing Analysis (STA) tool. It is the document that confirms whether a digital circuit meets its performance requirements. By meticulously analysing a timing report, a design engineer can identify and debug timing violations, which are paths where signals are arriving either too late or too early, potentially causing a functional failure in the final silicon.

APPLICATION:

- Identify the key sections within a timing report.
- Understand the purpose of different attributes and their values.
- Locate information about path endpoints / Begin Points
- Differentiate between setup and hold violations.

Step 1: Invoke OpenSTA in simple Directory

```
sta
```

Step 2: Read Library (STD CELLS LIB);

```
read_liberty osu018_stdcells.lib
```

```
OpenSTA> read_liberty osu018_stdcells.lib
```

Step 3: Read Netlist(.v)

```
read_verilog d_ff_synth.v
```

```
OpenSTA> read_verilog d_ff_synth.v
```

Step 4: Linking design

```
Link_design d_ff
```

```
OpenSTA> link_design d_ff
```

Step 5: Read SDC

```
read_sdc d_ff.sdc
```

```
OpenSTA> read_sdc d_ff.sdc
```

Step 5: Timing Report

report_checks

OpenSTA> report_checks

```
Startpoint: _4_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: q (output port clocked by clk)
Path Group: clk
Path Type: max

Delay    Time    Description
-----
0.00     0.00    clock clk (rise edge)
0.00     0.00    clock network delay (ideal)
0.00     0.00    ^ _4_/CLK (DFFPOSX1)
3.68     3.68    ^ _4_/Q (DFFPOSX1)
0.00     3.68    ^ q (out)
0.00     3.68    data arrival time

5.00     5.00    clock clk (rise edge)
0.00     5.00    clock network delay (ideal)
0.00     5.00    clock reconvergence pessimism
-5.00     0.00    output external delay
0.00     0.00    data required time

0.00     0.00    data required time
-3.68     0.00    data arrival time

-3.68    -3.68    slack (VIOLATED)
```

- Startpoint : The clock pin (f1:CLK) where the timing path begins
- Endpoint : The signal or port (out) where the timing path ends.
- Slack : The difference between required time and arrival time, showing timing margin.
- Arrival time : The actual time data takes to propagate from start to end
- Required time: The latest allowable time for data to arrive at the endpoint
- Violated: It means the timing requirement is not met because the data arrives too late.
- Why it's violated: The arrival time (30.724) is greater than the required time (20.000), so the slack is negative (-10.724), causing the violation

Note : You can generate timing reports using the following commands:

- Use `report_checks -path_delay max` to create the **setup timing report**, which checks if data arrives on time before the clock edge.
- Use `report_checks -path_delay min` to create the **hold timing report**, which checks if data is stable long enough after the clock edge.

OBSERVATION:

- The design is read and timing analysis is performed without file errors.
- The timing path is analyzed from f1:CLK to out.
- The arrival time is greater than the required time, indicating a timing violation.
- The slack is negative, meaning the data arrives too late at the endpoint.
- The setup timing check has failed for this path.

CONCLUSION:

The timing report from OpenSTA shows that the design's data path from clock f1:CLK to output out fails the setup timing check. The arrival time of 30.724 exceeds the required time of 20.000, resulting in a negative slack of -10.724 and causing a violation. This indicates that the circuit may not operate correctly at the target clock frequency, and timing optimization or redesign is necessary to ensure proper functionality.